

Evaluating the Reliability and Practical Viability of CNN-Based Early-Stage Plant Disease Detection Under Simulated Real-World Conditions

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Abstract

Convolutional neural networks achieve near-perfect accuracy on curated plant disease image datasets, yet performance falls sharply when the same models encounter real field imagery. Existing work documents this lab-to-field gap and catalogues its causes qualitatively without quantifying how much each capture condition contributes. This paper measures that contribution directly. An EfficientNet-B0 classifier was trained by transfer learning from ImageNet on a three-class tomato subset of PlantVillage (Early Blight, Late Blight, Healthy), then frozen. The frozen model was evaluated once against a clean 677-image test set and again against twelve synthetically degraded versions of that same test set, comprising four degradation conditions (brightness and contrast variation, Gaussian blur, Gaussian noise, and resolution reduction) at three severity levels each. Freezing the weights isolates image condition as the sole experimental variable. Clean accuracy was 94.98%, with a macro-averaged F1-score of 94.55%. Robustness varied by condition: accuracy remained at 91.43% under severe Gaussian noise but fell to 49.04% under severe Gaussian blur, a decline of 45.94 percentage points. Blur also produced a cliff-shaped rather than gradual decline, offering no early warning before failure. Across the three most damaging conditions macro recall fell faster than macro precision, indicating that degradation drives the model toward missed disease rather than false alarms, the costlier error in agriculture. A parallel model trained from random initialization showed that the transfer-learning advantage, 19.65 points on clean data, narrowed under noise and reversed entirely under severe blur and severe brightness and contrast degradation. Accuracy claims made on curated data therefore overstate field reliability, and architecture choice should reflect the capture conditions a deployment will face.

Keywords: plant disease detection; convolutional neural networks; corruption robustness; covariate shift; transfer learning; EfficientNet; precision agriculture

1. Introduction

Population growth has raised global demand for food faster than agricultural output can reliably meet it. Over 800 million people lack sufficient food, while at least 10% of global food production is lost to plant disease each year (Strange & Scott, 2005). This threat falls unevenly: in eastern Africa, where roughly 70% of the population is employed in agriculture (Omara et al., 2023, citing FAO, 2010), smallholder farmers growing subsistence crops are especially exposed, with reports indicating 35–40% of annual yield lost to disease in the region (Omara et al., 2023).

Disease detection in agricultural settings is commonly conducted through visual assessment, a process subject to human error and bias, while laboratory confirmation, though more reliable, is time-consuming and resource-intensive (Barbedo, 2016). These basic requirements are frequently undermined further by a shortage of trained personnel and diagnostic facilities (Strange & Scott, 2005), a gap visible even in modern deployed

systems, where one recent agricultural advisory tool reported diagnostic delays of five to seven days for farmers awaiting expert feedback (Omara et al., 2023). Fast, accurate identification remains vital to preserving yield (Strange & Scott, 2005), a requirement traditional approaches are structurally ill-equipped to meet consistently.

The rise of smartphone use has opened new possibilities for economically and practically viable disease detection using CNN architectures, a promise that notably excludes specialized hardware fixes such as engineered illuminant sets or polarization systems, impractical for deployment on typical consumer devices (Barbedo, 2016). Compared to traditional methods, CNN models have shown exceptional accuracy in classifying diseases: Mohanty et al. (2016) report a best-performing model reaching approximately 99% accuracy on curated laboratory images. However, cracks begin appearing when the same technology is applied to real-field conditions: the same study found accuracy collapsing to just 31.4% and 31.7% when evaluated against small, real-world image datasets sourced from the internet. Beyond the technical question of whether these models remain accurate under such conditions lies an economic one: whether a lower-cost, imperfect model still represents a net improvement over the status quo of manual, expertise-dependent detection.

This paper investigates both questions through a controlled experiment: a pretrained CNN classifier is evaluated once against a clean, curated test set, then repeatedly against systematically degraded versions of that same test set, with its weights held fixed throughout. Specifically, this paper asks: to what extent can CNN-based image classification systems reliably detect early-stage plant diseases under simulated real-world conditions, and what practical limitations affect their viability for agricultural deployment? In doing so, it extends Barbedo's (2016) call to study the influence of image capture conditions systematically, rather than treat it as noise to be eliminated, into a quantified, architecture-specific evaluation of a modern CNN. The remainder of this paper proceeds as follows: Section 2 reviews related work on CNN-based plant disease detection and the lab-to-field performance gap; Section 3 details the model architecture, dataset, and synthetic degradation methodology; Section 4 presents results from clean and degraded evaluations; and Section 5 discusses the economic implications of these findings for real-world agricultural deployment.

2. Related Work

2.1 CNNs in Plant Disease Detection

CNNs have become the standard architecture for leaf disease detection for two related but distinct reasons: they are technically effective, and they are practical to deploy. On the technical side, CNNs have supplanted the manual feature engineering required by traditional machine learning, allowing models to learn patterns directly from data (Sun et al., 2022). On the deployment side, Majdalawieh et al. (2025) find that CNNs remain the most cost-effective architecture for plant disease identification, even as alternatives like Vision Transformers offer superior accuracy at a computational cost that limits their practicality.

2.2 The Lab-to-Field Performance Gap

Despite strong laboratory performance, CNN-based classifiers exhibit a well-documented drop in accuracy when tested under real-world field conditions. Mohanty et al. (2016) report that while their best-performing model achieved approximately 99% accuracy on curated PlantVillage images, accuracy fell to just 31.4% and 31.7% when evaluated against two small real-world image datasets sourced from the internet. This is not an isolated result: Picon et al. (2019) similarly report that although their crop-conditional CNN architecture

achieved high performance on curated datasets, performance dropped substantially when evaluated on real, cell-phone-acquired field images. The pattern holds beyond these two studies as well: Majdalawieh et al.'s (2025) systematic review of 150 studies found that despite frequently high reported accuracies, the models evaluated were largely tested under narrow conditions and rarely assessed across the range of species, climates, or pest pressures they would encounter in practice. Taken together, these findings indicate that the lab-to-field performance gap is not an artifact of any single dataset or architecture, but a structural limitation of how plant disease classifiers are typically trained and evaluated.

2.3 Existing Deployment Attempts

Beyond controlled experimentation, some work has moved toward practical deployment, though the specific approaches taken reveal differing assumptions about the infrastructure available to end users. Sun et al. (2022) built their lightweight FL-EfficientNet architecture into a prototype web application, demonstrating deployment feasibility but implicitly assuming reliable internet connectivity. Omara et al. (2023) take the opposite approach: their MobileNetV3-based cassava disease classifier, built on Google's CropNet and deployed via TensorFlow Lite, was designed specifically for offline, on-device use, reflecting the reality that many smallholder farmers in the regions such tools are intended to serve lack consistent internet access. This contrast, a cloud-connected prototype versus an offline-first deployment, reflects the same infrastructure-cost tradeoff this paper's economic analysis later addresses directly.

2.4 The Gap This Paper Addresses

While Barbedo (2016) provides a comprehensive taxonomy of the specific factors that contribute to the lab-to-field performance gap (including illumination variability, capture angle, sensor noise, and resolution constraints), this taxonomy remains primarily descriptive: it catalogs the sources of covariate shift without quantifying their individual effect on a modern CNN architecture's classification performance. The methodology for such quantification is, however, well established in general computer vision. Hendrycks and Dietterich (2019) introduced ImageNet-C, a benchmark that evaluates classifiers against fifteen synthetic corruption types applied at five severity levels each, and showed that architectures with comparable clean accuracy can differ substantially in their robustness to corruption. Four of those corruption types (Gaussian noise, defocus blur, brightness, and contrast) correspond closely to the capture conditions Barbedo identifies as most problematic in agricultural imaging, and a fifth, pixelation, approximates the effect of resolution reduction.

This severity-graded protocol has not, to the author's knowledge, been applied to plant disease classification, where three features of the task make the outcome difficult to predict from general-purpose benchmarks. First, the discriminative features are fine-grained lesion textures rather than object shape or silhouette, so blur and resolution loss may prove considerably more damaging than they are for general object recognition. Second, the cost of errors is asymmetric: a diseased leaf classified as healthy carries substantially higher real-world cost than the reverse, so aggregate accuracy is an incomplete measure of deployment viability. Third, several of the relevant degradations are deliberate engineering choices rather than incidental noise, since reduced resolution and image compression are precisely the levers a low-cost or bandwidth-constrained deployment would pull.

This paper therefore applies a severity-graded corruption protocol in the manner of Hendrycks and Dietterich (2019) to a three-class tomato disease task, measuring how much accuracy EfficientNet-B0 loses under each condition as severity increases, and interpreting the result against the cost structure of agricultural deployment. The methodology is detailed in Section 3.

3. Methodology

3.1 Model Architecture

EfficientNet-B0, pretrained on ImageNet, was selected as the primary classification architecture, implemented via transfer learning rather than training from random initialization. EfficientNet improves model efficiency by jointly scaling network depth, width, and input resolution (Sun et al., 2022), making it a natural fit for a task where deployment efficiency matters as much as raw classification accuracy.

This choice reflects a pattern already established in the plant disease detection literature. Majdalawieh et al. (2025) observe that the field's continued preference for CNN architectures (and its sustained reliance on transfer learning specifically) reflects a deliberate effort to optimize feature extraction for domain-specific agricultural tasks while reducing computational demands. Mohanty et al. (2016) provide direct empirical support for that preference, finding that transfer learning from ImageNet consistently outperformed training from scratch across every architecture and dataset configuration tested. Rather than accepting this finding on the strength of prior literature alone, this paper verifies it directly: a second model, identical in architecture but trained from random initialization, was evaluated in parallel as a comparison baseline (see Section 4.3).

EfficientNet-based architectures have also demonstrated strong classification performance elsewhere in this literature: Sun et al. (2022), for instance, report 99.7% accuracy using an EfficientNet variant on the NPDD dataset, spanning five crops and ten diseases. This figure is not directly comparable to the results reported here, given the difference in dataset scope and the narrower three-class, single-crop task addressed in this paper; it is included as evidence of the architecture family's general capability rather than as a benchmark target. More directly relevant to this paper's aims is deployability: Sun et al. (2022) built their lightweight EfficientNet variant into a working prototype web interface, explicitly treating deployment feasibility as a design consideration alongside accuracy. Mohanty et al. (2016) reach a related conclusion from a different angle: although training a CNN is computationally expensive, typically requiring several hours on high-performance GPU hardware, the resulting inference is fast, under one second on a CPU, making smartphone-based deployment realistic without specialized hardware. Together, these findings support EfficientNet-B0 as a defensible choice not only for its classification performance, but for its compatibility with the low-cost deployment context this paper's later economic analysis addresses.

The model was trained on 128×128 input images, smaller than EfficientNet-B0's canonical 224×224 configuration. This choice was driven primarily by training-time constraints on the local hardware used for this project, rather than an a priori design decision. The resulting reduction in computational footprint is not incidental to this paper's broader interest in low-cost deployment viability, however, and is revisited briefly in the discussion of practical constraints.

3.2 Dataset

The dataset used is a subset of PlantVillage, restricted to tomato leaf images. Majdalawieh et al.'s (2025) systematic review of machine learning methods for crop disease detection reports that PlantVillage is the dominant dataset used across the reviewed literature. Building on a widely-used dataset keeps this paper's findings directly comparable to prior work, rather than introducing dataset-specific idiosyncrasies that would complicate comparison.

The decision to restrict scope to a single crop, rather than pursue a multi-crop dataset, is similarly grounded in the reviewed literature. Among the studies covered by Majdalawieh et al. (2025), tomato was the most

frequently studied crop, appearing in 31 studies, ahead of rice (18) and grape and potato (8 each). Concentrating on a single, well-represented crop allows this paper to prioritize depth of robustness testing (the twelve degradation conditions detailed in Section 3.4) over breadth of dataset coverage, which was judged the more valuable contribution given the scope of a single research project.

Three classes were used: Tomato Early Blight, Tomato Late Blight, and Tomato Healthy. The dataset was split 70/15/15 into training, validation, and test sets prior to any further processing, with degradation applied only to the test split (Section 3.6). The resulting test set comprises 677 images across the three classes: 150 Tomato Early Blight, 287 Tomato Late Blight, and 240 Tomato Healthy.

3.3 Covariate Shift Framing

The experimental design adopted here (evaluating a single, frozen model against systematically varied image conditions) reflects a specific reframing of the well-documented gap between laboratory and field performance. Rather than treating variation in image capture conditions as noise to be eliminated through stricter acquisition protocols, this paper treats it as a variable to be measured directly. This follows Barbedo (2016), who argues that a more productive research strategy than attempting to eliminate capture-condition variability entirely is to systematically study how such factors affect model performance, and to design methods that account for them accordingly.

Barbedo further situates this problem within the concept of covariate shift (Sugiyama et al., 2008), a mismatch between the distribution of data a model is trained on and the distribution it encounters at deployment, noting that this framing is particularly relevant to plant disease identification, where symptom presentation is known to vary geographically. Where Barbedo's treatment of covariate shift remains largely descriptive, cataloguing the range of conditions that contribute to it, this paper's synthetic degradation testing (Section 3.4) operationalizes the concept directly: each degradation condition constitutes a controlled, isolated instance of covariate shift, with its effect on classification performance measured quantitatively rather than inferred qualitatively.

3.4 Synthetic Degradation Conditions

Four degradation conditions were applied to the clean test set, each at three severity levels (mild, moderate, severe), producing twelve degraded test sets in total. Severity levels were defined qualitatively before being mapped to specific parameters: mild approximates a plausible best-case field condition, moderate a typical field condition, and severe a worst-case but still realistic condition. Each condition was selected to approximate a specific, literature-documented source of image variability rather than degradation in the abstract.

Brightness and contrast variation (mild: $\alpha=0.9$, $\beta=-10$; moderate: $\alpha=0.75$, $\beta=-30$; severe: $\alpha=0.6$, $\beta=-50$) approximates inconsistent field lighting. Illumination inconsistency has been identified as a persistent obstacle across multiple, unrelated plant pathology imaging studies (spanning citrus canker severity assessment, seagrass injury analysis, and citrus disease identification; Barbedo, 2016), suggesting a cross-domain problem rather than an isolated finding specific to one crop or imaging setup. Barbedo further notes that specular reflection and the simultaneous presence of shadow and direct illumination are especially difficult to control, and that automated image capture makes such conditions more likely than careful manual capture.

Gaussian blur (mild: kernel size 3, $\sigma=1$; moderate: kernel size 7, $\sigma=3$; severe: kernel size 15, $\sigma=6$) approximates inconsistent capture angle and focus. Barbedo (2016) identifies capture angle and leaf orientation as a further

source of image variability, noting that slanted capture angles can cause partial loss of focus; this paper treats angle-induced blur as subsumed within the Gaussian blur condition rather than isolating it separately.

Gaussian noise (mild: $\sigma=5$; moderate: $\sigma=15$; severe: $\sigma=30$) approximates the image-quality limitations of low-cost consumer devices. Barbedo (2016) notes that while mobile phone cameras can produce reasonable image quality under good lighting, the very small symptoms characteristic of early-stage disease require high optical quality and resolution to be reliably captured, a constraint of particular relevance given this paper's focus on early-stage detection specifically, rather than disease identification at a more visually advanced and distinct stage.

Resolution reduction (mild: 50% scale; moderate: 25% scale; severe: 10% scale) approximates both hardware and bandwidth constraints relevant to low-cost deployment. Barbedo (2016) identifies limited computational resources as a historical barrier motivating reduced image resolution; while general compute capacity has increased substantially since 2016, this constraint remains directly relevant to the lowest-cost devices available to smallholder farmers in resource-constrained regions, reframing an originally hardware-oriented concern as a deployment-cost concern. Barbedo also notes that image compression, often necessary to manage bandwidth and storage costs in cloud-based deployment, results in data loss that can compromise the visibility of fine disease symptoms, a tension directly relevant to this paper's economic analysis (Section 5.4).

Each condition was applied independently to isolate its individual effect, rather than in combination. This is a conservative simplification relative to real field conditions, where multiple degradations typically co-occur simultaneously; the implications of this simplification are revisited in Section 6 (Limitations).

3.5 Evaluation Metrics

Because accuracy alone can obscure how a model fails (and not all failures carry equal cost in an agricultural deployment context), this paper reports four complementary metrics: accuracy, precision, recall, and F1-score, with the latter three macro-averaged across the three classes. Macro-averaging assigns equal weight to each class regardless of how many test images it contains, ensuring strong performance on a well-represented class cannot mask weak performance on a smaller one.

A confusion matrix is reported alongside these aggregate metrics for each evaluation, making specific misclassification patterns visible rather than inferred from summary statistics alone. This distinction matters directly for this paper's later economic analysis: a false negative, a diseased leaf classified as healthy, carries substantially higher real-world cost than a false positive, since it risks delayed intervention and continued crop loss, whereas a false positive typically results only in unnecessary but reversible precautionary action. Reporting the confusion matrix surfaces this asymmetry in a way a single macro-averaged accuracy figure cannot.

3.6 Evaluation Design: Train Once, Test Many

The model's weights were frozen immediately after Phase 1 training and never subsequently updated. Every evaluation reported in this paper, the clean baseline (Phase 2) and each of the twelve degraded conditions (Phase 4), was performed using this same frozen model, evaluated repeatedly against different test inputs rather than retrained for each condition individually.

This design choice is central to the paper's core claim. Had the model been retrained, or its weights adjusted, for each degradation condition, any resulting difference in performance could not be attributed cleanly to the

image conditions themselves; it could equally reflect differences in how each version of the model happened to be trained. Holding the model constant isolates image condition as the sole variable under test.

Degradation was applied exclusively to the test split; training and validation data were left untouched throughout, preventing data leakage between what the model learned from and what it was subsequently evaluated against. Degraded images retained the same filenames as their clean counterparts, enabling direct image-level comparison during analysis, for instance identifying whether an image correctly classified under clean conditions remained correctly classified under a given degraded condition, or whether misclassification concentrated among images that were already borderline cases in the clean evaluation. Three severity levels per condition were judged sufficient to reveal the shape of each degradation curve (whether performance decline is gradual or occurs sharply past a threshold) without expanding Phase 3 image generation into an open-ended project in its own right.

4. Results

4.1 Clean Baseline Performance

On the clean, undegraded test set, the transfer-learning model achieved an overall accuracy of 94.98% and a macro-averaged F1-score of 94.55% across the three classes. Table 1 reports the per-class breakdown; Figure 1 presents the corresponding confusion matrix. Class-level performance was not uniform: the Healthy class was predicted with perfect precision (100%) but recall of 96.67%, while Early Blight recorded the lowest recall of the three classes (88.67%), indicating it was the class most frequently confused with another. These results establish a clean-condition performance ceiling against which the degraded evaluations in Section 4.2 are compared.

Class	Precision	Recall	F1	Support
Tomato Early Blight	93.66%	88.67%	91.10%	150
Tomato Late Blight	91.75%	96.86%	94.24%	287
Tomato Healthy	100.00%	96.67%	98.31%	240

Table 1. Per-class performance on the clean test set (EfficientNet-B0, transfer learning).

Figure 1. Confusion matrix for the clean, undegraded test set. Cell values give image counts, with the percentage of each true class beneath. Overall accuracy 94.98% (643 of 677).

4.2 Performance Under Synthetic Degradation

Table 2 reports accuracy and macro-averaged precision, recall, and F1-score for the model evaluated against each of the twelve degraded test sets described in Section 3.4. Figure 2 presents these results as degradation curves, plotting accuracy against severity for each condition alongside the clean baseline established in Section 4.1.

Condition	Severity	Accuracy	Macro Precision	Macro Recall	Macro F1
Clean baseline	N/A	94.98%	N/A	N/A	94.55%

Condition	Severity	Accuracy	Macro Precision	Macro Recall	Macro F1
Brightness/Contrast	Mild	93.94%	94.21%	93.09%	93.57%
Brightness/Contrast	Moderate	86.85%	89.25%	85.91%	86.83%
Brightness/Contrast	Severe	70.31%	79.19%	67.68%	69.08%
Gaussian Blur	Mild	88.33%	90.90%	87.71%	88.57%
Gaussian Blur	Moderate	62.33%	74.21%	64.68%	54.75%
Gaussian Blur	Severe	49.04%	66.86%	54.65%	41.16%
Gaussian Noise	Mild	95.13%	95.49%	94.08%	94.70%
Gaussian Noise	Moderate	94.24%	94.52%	92.95%	93.63%
Gaussian Noise	Severe	91.43%	91.38%	89.62%	90.36%
Resolution Reduction	Mild	94.83%	95.02%	93.84%	94.37%
Resolution Reduction	Moderate	88.04%	89.06%	85.92%	87.09%
Resolution Reduction	Severe	65.29%	66.93%	63.89%	64.83%

Table 2. Accuracy and macro-averaged precision, recall, and F1-score for the frozen model under each degradation condition and severity level.

Figure 2. Top-1 accuracy against degradation severity for each of the four conditions, plotted from the clean baseline. The frozen model was evaluated on the same 677-image test set at every point.

Across all four conditions, accuracy declined monotonically from mild to severe severity, with no reversals, consistent with the severity parameters described in Section 3.4 producing meaningfully distinct levels of image corruption rather than arbitrary variation. The magnitude of decline, however, varied substantially by condition. Gaussian blur produced the largest drop in performance, falling from 88.33% accuracy at mild severity to 49.04% at severe severity, a 45.94-percentage-point decline relative to the clean baseline, the largest single-condition decline observed in this study. Resolution reduction and brightness/contrast variation produced comparable, moderate declines, reaching 65.29% and 70.31% accuracy respectively at severe severity. Gaussian noise proved the condition to which the model was most robust: even at severe severity ($\sigma=30$), accuracy remained at 91.43%, only 3.55 percentage points below the clean baseline.

A further pattern is visible in the precision-recall breakdown. Across the three more damaging conditions (blur, resolution reduction, and brightness/contrast), macro-averaged recall declined more steeply than macro-averaged precision as severity increased. Under severe blur, for instance, macro precision remained at 66.86% while macro recall fell to 54.65%; a similar asymmetry is visible under severe brightness/contrast variation (79.19% precision vs. 67.68% recall). This indicates that as image quality degrades, the model increasingly fails by missing true instances of disease rather than by raising false alarms.

4.3 Transfer Learning vs. From-Scratch Comparison

To test directly whether the transfer-learning advantage established in the literature (Mohanty et al., 2016; Section 3.1) holds under this paper's degraded test conditions, an identically-architected model trained from random initialization was evaluated against the same clean and degraded test sets used in Sections 4.1 and 4.2.

On the clean test set, the transfer-learning model substantially outperformed the from-scratch model, achieving 94.98% accuracy and 94.55% macro F1 compared to 75.33% accuracy and 74.07% macro F1 for the from-scratch model, a 19.65-point accuracy gap. This result is consistent with Mohanty et al.'s (2016) finding that transfer learning from ImageNet consistently outperforms training from scratch.

Table 3 reports accuracy for both models across all twelve degraded conditions, alongside the resulting gap. The transfer-learning advantage was not uniform across degradation types. Under resolution reduction, the gap remained broadly stable across all three severities (19.35, 15.07, and 18.61 percentage points), indicating the transfer-learning advantage persists largely undiminished under this condition. Under Gaussian noise, the gap narrowed substantially with increasing severity (18.76, 8.72, and 7.53 percentage points), driven primarily by the from-scratch model's accuracy *increasing* under noise rather than declining, reaching 85.52% at moderate severity, higher than its own clean-baseline accuracy of 75.33%.

Under two conditions, the transfer-learning advantage reversed entirely at severe severity. Under severe brightness/contrast degradation, the from-scratch model slightly outperformed the transfer-learning model (71.05% vs. 70.31%, a -0.74 -point gap). Under severe Gaussian blur, this reversal was considerably larger: the from-scratch model achieved 59.68% accuracy compared to the transfer-learning model's 49.04%, a -10.64 -point gap in the from-scratch model's favor. Notably, blur also produced the widest transfer-learning advantage observed in this study at mild severity (30.58 percentage points), before collapsing and reversing by severe severity, the largest single-condition swing in relative model performance recorded across all evaluations.

Condition	Severity	Transfer Acc.	From-Scratch Acc.	Gap (pts)
Clean baseline	N/A	94.98%	75.33%	19.65
Brightness/Contrast	Mild	93.94%	73.71%	20.23
Brightness/Contrast	Moderate	86.85%	71.94%	14.91
Brightness/Contrast	Severe	70.31%	71.05%	-0.74
Gaussian Blur	Mild	88.33%	57.75%	30.58
Gaussian Blur	Moderate	62.33%	53.18%	9.15
Gaussian Blur	Severe	49.04%	59.68%	-10.64
Gaussian Noise	Mild	95.13%	76.37%	18.76
Gaussian Noise	Moderate	94.24%	85.52%	8.72
Gaussian Noise	Severe	91.43%	83.90%	7.53
Resolution Reduction	Mild	94.83%	75.48%	19.35
Resolution Reduction	Moderate	88.04%	72.97%	15.07
Resolution Reduction	Severe	65.29%	46.68%	18.61

Table 3. Clean and degraded accuracy for the transfer-learning and from-scratch models, with the resulting gap in percentage points. Negative values indicate the from-scratch model outperformed the transfer-learning model.

These reversals are notable but should be interpreted cautiously: each condition/severity combination reflects a single evaluation run rather than an average across repeated trials, and no statistical significance testing was performed to establish whether these gap reversals reflect a robust underlying pattern or single-run variance. This limitation is discussed further in Section 6.

5. Discussion

5.1 Interpreting the Shape of Degradation

The degradation results reported in Section 4.2 reveal not only that certain conditions harm performance more than others, but that the *shape* of decline differs meaningfully across conditions, a distinction with direct implications for real-world reliability. Gaussian noise produced a shallow, gradual decline (94.98% clean to 91.43% at severe severity), suggesting a deployment context dominated by sensor noise would degrade predictably and gently. Gaussian blur, by contrast, produced a pronounced cliff: performance held reasonably well at mild severity (88.33%) before collapsing sharply by moderate severity (62.33%) and further by severe severity (49.04%). This distinction matters practically. A gradual decline offers an implicit early-warning signal: accuracy visibly worsens before it becomes unreliable. A cliff-shaped decline offers no such warning: the model may appear reliable under mild-to-moderate blur, then fail sharply once a threshold is crossed, with no visible signal to a non-expert user that this threshold has been reached. Of the four conditions tested, blur therefore represents the clearest reliability risk for real-world deployment, independent of its average accuracy cost.

5.2 The Cost of Being Wrong

Raw accuracy alone understates the practical cost of degradation, because not all classification errors carry equal consequence for a farmer. As established in Section 4.2, macro-averaged recall declined more steeply than macro-averaged precision across the three most damaging conditions: under severe blur, for instance, precision remained at 66.86% while recall fell to 54.65%. In practical terms, this means the model's failures under degradation are increasingly failures of omission (misclassifying a diseased leaf as healthy) rather than failures of false alarm. This asymmetry matters economically: a false negative allows disease to progress undetected, risking continued crop loss and delayed intervention, while a false positive typically results only in unnecessary but reversible precautionary action, such as an unneeded pesticide application. Given that recall degrades faster than precision as field conditions worsen, accuracy alone is an incomplete metric for evaluating deployment viability in this context; recall under realistic conditions is arguably the more economically relevant figure.

5.3 The Training-Cost Tradeoff Revisited

The transfer-learning versus from-scratch comparison reported in Section 4.3 complicates the standard advice, supported by Mohanty et al. (2016) and reflected in this paper's own clean-baseline results, that transfer learning is unambiguously the superior choice. At the level of inference, this comparison carries no economic weight: both models share an identical architecture, and therefore an identical computational cost and latency at deployment time, regardless of how they were trained. The real economic distinction lies upstream, in

training. Majdalawieh et al. (2025) note that transfer learning reduces computational and data requirements by leveraging pretrained feature representations rather than learning them from a randomly initialized state, a meaningful advantage for a developer with limited access to large labeled datasets, a constraint directly connected to the shortage of trained diagnostic personnel established in Section 1 (Strange & Scott, 2005).

However, this paper's degradation results suggest the accuracy benefit of transfer learning is not guaranteed under all field conditions. While the transfer-learning model held a substantial advantage on clean data and under resolution reduction, its advantage narrowed considerably under Gaussian noise and reversed entirely under severe blur and severe brightness/contrast degradation. This suggests the decision to use transfer learning should be informed by the specific degradation conditions a deployment is likely to face, rather than treated as a universally optimal default. A developer with reason to expect their target environment involves inconsistent focus or lighting, for instance, may find a from-scratch model a more competitive choice than the literature's general guidance would suggest, though, given this finding rests on a single evaluation run per condition (Section 4.3), it warrants confirmation through repeated trials before being treated as a firm deployment recommendation.

5.4 The Cheap-vs-Accurate Deployment Tradeoff

Taken together, these findings inform this paper's central economic question: given that all models tested degrade under field-like conditions to some degree, is it rational for a resource-constrained deployment (such as a tool intended for smallholder farmers) to favor a cheaper, lower-accuracy model over a more accurate but more resource-intensive one? Majdalawieh et al. (2025) find that CNNs remain the most cost-effective architecture choice across the reviewed literature specifically because more accurate alternatives, such as Vision Transformers, demand computational resources that limit real-world practicality, the same underlying tradeoff this paper's transfer-versus-scratch comparison surfaces at a finer grain. Existing deployment precedent supports the practical viability of this cost-conscious approach: Omara et al. (2023) demonstrate a working, offline, on-device disease classifier deployed via TensorFlow Lite specifically for hardware- and connectivity-constrained agricultural settings, achieving meaningful accuracy without cloud dependency.

The resolution-reduction results reported in Section 4.2 offer a further, direct illustration of this tradeoff. Barbedo (2016) notes that image compression, often necessary to manage the bandwidth and storage costs of cloud-based deployment, results in data loss that can compromise the visibility of fine disease symptoms. This paper's own results quantify that tension directly: reducing image resolution to 25% of its original scale cost 6.94 percentage points of accuracy (94.98% to 88.04%), while reducing it further to 10% scale cost 29.69 percentage points (94.98% to 65.29%). A deployment prioritizing minimal bandwidth or storage cost would need to weigh these concrete accuracy losses against the infrastructure savings such compression provides, a tradeoff this paper's results make quantifiable rather than merely qualitative.

5.5 Deployment Reliability: Calibration and the Unknown-Class Problem

One limitation of the evaluation framework used throughout this paper is worth surfacing explicitly in the context of real deployment. The model evaluated here is a closed-set, three-class classifier: it will always output one of Tomato Early Blight, Tomato Late Blight, or Tomato Healthy, regardless of whether the input image genuinely resembles any of these three categories. An image of an unfamiliar disease, a different crop entirely, or an image degraded beyond meaningful recognition will still receive a confident-looking prediction, because the model has no mechanism to express genuine uncertainty or decline to classify. The confidence scores reported throughout this paper, drawn directly from softmax output, measure the model's relative certainty among its three known classes, not a calibrated estimate of the probability that a prediction is

correct, nor an indication of whether the input belongs to a known category at all. A production deployment intended for real farmers would benefit from an explicit "unknown" or low-confidence rejection path, alongside confidence calibration, so that genuinely unfamiliar or severely degraded inputs are flagged for human review rather than confidently misclassified.

This paper's own results suggest where such a safeguard would matter most: severe Gaussian blur produced both the steepest accuracy decline and the widest reversal in relative model performance observed in this study, making it the clearest single priority for future data collection: specifically, collecting or synthetically augmenting training data with realistic out-of-focus field images, rather than distributing further collection effort evenly across all degradation types.

6. Limitations

Several limitations constrain the generalizability of this paper's findings.

Scope of dataset and classes. This paper evaluates a single crop (tomato) across three classes (Early Blight, Late Blight, Healthy), a deliberate scope decision made to prioritize depth of robustness testing over breadth of dataset coverage (Section 3.2). This narrow scope limits the extent to which these findings generalize to other crops, diseases, or classification granularities; a model's robustness to a given degradation condition may differ substantially for a different crop or disease presentation. Additionally, the PlantVillage images used throughout this study were collected under controlled laboratory conditions; Mohanty et al. (2016) note that models trained on such images risk learning dataset-specific visual artifacts, such as a uniform background, rather than genuine disease morphology, a risk this paper's synthetic degradation testing does not directly rule out, since the underlying images being degraded remain lab-sourced rather than genuinely field-captured.

Synthetic rather than real degradation. The four degradation conditions applied in this study (Section 3.4) were synthetically generated approximations of real-world capture conditions, applied independently rather than in combination. Real field conditions frequently involve multiple degradations occurring simultaneously: for instance, an image captured in poor lighting on an inexpensive camera may exhibit blur, noise, and brightness variation at once. By isolating each condition, this paper measures the individual effect of each degradation type clearly, but likely understates the compounded performance cost a model would face under genuinely realistic field conditions. Relatedly, this paper does not include a validation set of genuinely field-captured images, a stretch goal considered but not pursued given project scope; synthetic degradation, while grounded in the literature-documented sources of variability it approximates (Barbedo, 2016), remains an approximation rather than a substitute for real-world field validation.

Single-run evaluation. Each condition/severity combination reported in Section 4, including the transfer-versus-scratch comparison in Section 4.3, reflects a single evaluation run rather than an average across repeated trials. This is a particular concern for the reversals observed between the transfer-learning and from-scratch models under severe blur and severe brightness/contrast degradation (Section 4.3): while these results are internally consistent and evaluated against identical test data, no statistical significance testing was performed, and it remains possible that some finer-grained differences reported reflect single-run variance rather than a robust underlying pattern, though this is less likely for the large-magnitude reversals given their consistency.

No mechanism for unknown or out-of-distribution inputs. As discussed in Section 5.5, the model evaluated throughout this paper is a closed-set, three-class classifier with no calibrated confidence estimation and no rejection pathway for inputs that do not genuinely resemble any of its three known classes. This is an appropriate design for a controlled research evaluation, where the goal is to measure classification performance under known conditions, but it is a meaningful limitation for any discussion of production deployment, where a model will inevitably encounter inputs (different crops, unfamiliar diseases, severely corrupted images) outside its training distribution.

Untested conditions. This paper's four degradation conditions do not exhaustively cover the range of capture-condition variability documented in the literature (Section 3.4). Motion blur, distinct from the Gaussian blur tested here and arising specifically from camera movement during capture, was considered during experimental design but not included in the final evaluation. Combined-condition testing, approximating the co-occurrence of multiple degradations, was similarly considered but left for future work given project scope.

Economic analysis grounded in secondary data. The economic interpretation offered in Section 5 draws on accuracy and cost-effectiveness findings reported in the existing literature (e.g., Majdalawieh et al., 2025) rather than on primary cost data collected specifically for this study, such as measured training time, hardware pricing, or bandwidth costs specific to the deployment regions discussed. The qualitative direction of the tradeoffs discussed is well-supported by this paper's own accuracy results and by cited literature, but precise economic figures, such as an exact cost-per-percentage-point-of-accuracy tradeoff, would require dedicated economic data collection beyond this paper's scope.

7. Conclusion

This paper set out to evaluate the extent to which CNN-based image classification systems can reliably detect early-stage plant disease under simulated real-world conditions, and what practical limitations affect their deployment viability. Using EfficientNet-B0 with transfer learning, evaluated against a clean baseline and twelve synthetically degraded test conditions, this paper finds that model performance is not uniformly robust: while the model achieved 94.98% accuracy on clean, curated images, performance declined substantially under several degradation conditions, most severely under Gaussian blur, which produced a 45.94-percentage-point accuracy drop at severe severity. Notably, the model's failures under degradation skewed toward false negatives rather than false positives, a pattern with direct economic significance given the higher real-world cost of undetected disease relative to unnecessary precautionary action.

A comparison against an identically-architected model trained from scratch confirmed the well-established finding that transfer learning substantially outperforms training from random initialization on clean data, consistent with Mohanty et al. (2016). However, this advantage was not uniform under degradation: it persisted under resolution reduction, narrowed under Gaussian noise, and reversed entirely under severe blur and severe brightness/contrast conditions, with the from-scratch model outperforming the transfer-learning model in both cases. This finding complicates the assumption that transfer learning is a universally optimal choice, suggesting instead that architecture selection should be informed by the specific degradation conditions a deployment context is likely to face.

These findings carry direct implications for the economic question motivating this paper: whether a lower-cost, imperfect detection tool represents a net improvement over the status quo of manual, expertise-dependent disease identification for smallholder farmers. The evidence gathered here suggests a qualified yes,

with important caveats. CNN-based detection remains substantially more accurate than chance even under significant degradation, and existing deployment precedent (Omara et al., 2023) demonstrates the practical viability of low-cost, on-device implementations. However, this paper's results also demonstrate that accuracy claims made on curated laboratory data substantially overstate real-world reliability, and that the specific shape of a model's degradation curve, gradual versus cliff-like, carries reliability implications a single accuracy figure cannot capture.

Future work should prioritize three directions suggested directly by this paper's findings. First, given that severe blur produced both the steepest performance decline and the clearest transfer-learning reversal observed in this study, collecting or synthetically augmenting training data with realistic out-of-focus field images represents the most clearly indicated next data-collection priority. Second, repeated evaluation trials across degradation conditions would establish whether the transfer-versus-scratch reversals reported in Section 4.3 reflect a robust underlying pattern or single-run variance. Third, extending this evaluation framework to include a calibrated confidence mechanism and an explicit rejection pathway for out-of-distribution inputs, as discussed in Section 5.5, would meaningfully narrow the gap between this paper's research evaluation and a genuinely deployment-ready system.

Ultimately, this paper's central contribution is a quantified answer to a question the existing literature poses largely in qualitative terms: not simply that CNN-based plant disease classifiers lose accuracy under field conditions, but precisely how much, under which conditions, and with what economic consequence, directly extending Barbedo's (2016) call to study capture-condition variability systematically rather than treat it as noise to be eliminated.

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